Artificial Intelligence

BS (CS) Spring 2026

Lab 03 Tasks



Learning Objectives:

1. Agent Basis
2. Environment
3. Simple Reflex Agent
4. Model Based Agent
5. Goal Based Agent
6. Code Examples

Lab Tasks

Submission Instructions

1. Create a notebook file for all questions with name *ROLLNO_SEC_LAB3* **e.g. i23XXXX_A_LAB03**
2. Compress your **.ipynb** file along with output **screenshots** directly into a **.zip file**.
3. Now you have to submit this zipped file on Google Classroom under your section heading.
4. If you don't follow the above-mentioned submission instruction, you will be marked **zero.**

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

Questions

Q1. Design a smart robot that operates inside a warehouse. The robot is equipped with obstacle-detecting sensors and can move in four directions: forward, backward, left, and right. Its objective is to navigate the warehouse efficiently and reach a specified destination. Using a agent approach, the robot must make decisions based solely on its current percepts, reacting to obstacles in its immediate surroundings without remembering past moves or predicting future ones. Represent the warehouse as a 2D grid, where free spaces and obstacles are indicated appropriately. Write a program that enables the robot to detect obstacles, select its next move using condition–action rules, and print the sequence of moves taken to reach the goal or indicate if the destination is unreachable.

```
env=[ S, 0, 1, 0, 0 ]
[ 0, 0, 1, 0, 1 ]
[ 1, 0, 0, 0, 0 ]
[ 0, 1, 1, 0, 0 ]
[ 0, 0, 0, 1, G ]
```

`0` = free space

`1` = obstacle

`S` = start position

`G` = goal/destination

Q2. Imagine you are designing a warehouse inspection robot tasked with monitoring a warehouse floor represented as a 5×5 grid. Certain areas of the grid may be **damaged (x)** or contain **obstacles (#)**, while other areas are safe to traverse. The robot is equipped with sensors that allow it to detect whether a grid cell has already been inspected. Its actions include moving in four directions and inspecting the current cell. The robot traverse the grid untill the robot inspect all reachable areas of the warehouse without revisiting any cell**,** while avoiding obstacles. Design an agent for this task maintaining an internal model of the warehouse and uses it, along with its current percepts, to decide which action to take next.

Q3. Design a goal-based agent for a food delivery robot.

Task Description:

- Environment:
  - A 6x6 grid with Food Pickup Points (F).
- Percepts:
  - Current position and pending pickups.
- Actions:
  - Move and Pick.
- Goal:
  - Complete all food pickups.